

SAKSHI KRISHNA PATIL

Flat no. 2, Suhas App., Yashwantnagar, Pimpri, Pune 18

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Education

Pimpri Chinchwad College of Engineering

Bachelor of Science in Electronics and Telecommunication Engineering

Aug. 2020 – May 2024

Pune, Maharashtra

Certification / Coursework

- Data Structures
- Algorithms Analysis
- AIML
- Javascript
- Python
- C, C++
- MATLAB Onramp
- SQL

Experience

PHN Technology Pvt Ltd

Machine Learning and Data Science Intern

June 2024 Nov 2024

Pune, Maharashtra

- Developed and trained deep learning models using TensorFlow and Keras, achieving 92% accuracy on plant disease classification tasks.
- Developed supervised and deep learning models using Python, TensorFlow, and Keras to classify and analyze structured and unstructured datasets.
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) using Pandas, NumPy, Matplotlib, and Seaborn.
- Built and evaluated machine learning pipelines for predictive analytics tasks, improving model accuracy by 10% through hyperparameter tuning.
- Created visual dashboards using Plotly and Seaborn to present actionable insights to technical and non-technical stakeholders.
- Collaborated with cross-functional teams to apply AI/ML techniques in real-world applications such as plant disease detection and audio classification.

Oasis Infobyte

Web Development Intern

Dec 2023- Feb 2024

Pune, Maharashtra

- Identified and resolved 30+ bugs during testing phases, ensuring 100% cross-browser functionality.
- Built and deployed responsive web pages using HTML, CSS, JavaScript, and React with backend logic in Node.js and Python.
- Collaborated with team members using version control systems such as Git to organize modifications and assign tasks.
- Utilized Android Studio as a development environment in order to visualize the application in both iOS and Android.
- Collaborated with a team of 4 to implement REST APIs and streamline form handling, reducing response time by 40%.

Kamal Developers Engineers & Contractors

Graduate Engineer Trainee Intern

Jan 2025 April 2025

Pune, Maharashtra

- Participate in technical training sessions and onboarding programs to understand tools, systems, and project requirements.
- Perform basic data collection, analysis, and interpretation to support ongoing engineering tasks.
- Prepare and maintain documentation for designs, processes, and systems, adhering to company standards.
- Identify technical issues during development or testing and collaborate with team members to resolve them.
- Work closely with cross-functional teams including design, development, and quality assurance.

Projects

Smart blind stick using JQ650 Voice Sound Module | Embedded C, C, Sensors, JQ6500, Arduino

January 2022

- Designed and developed an assistive smart stick for the visually impaired using the JQ6500 Voice Sound Module and ultrasonic sensors to detect obstacles.
- Programmed the system in Embedded C to trigger voice alerts based on the direction of the detected object (e.g., "Object on Left").
- Integrated Arduino microcontroller to interface with sensors and sound modules, enabling real-time object detection and response.
- Optimized circuit design for compactness and portability, enhancing user comfort and ease of use.
- Conducted rigorous field testing and improved the voice feedback system for better clarity and responsiveness.
- The blind person can hear the beeping of the buzzer and manage to change the way. In this way, the person can easily find his way without getting injured.

Plant Disease Detection using AIML | Machine Learning, Neural Networks, CNN, OpenCV

November 2023

- Developed an AI-based system to detect and classify plant diseases using Convolutional Neural Networks (CNNs) and supervised learning algorithms.
- Built a custom dataset by collecting and preprocessing over 10,000+ leaf images representing healthy and diseased plants across various species.
- Achieved accuracy of 92% in disease classification through data augmentation, re-tuning, and model optimization.
- Utilized OpenCV for image processing and feature extraction, enhancing model input quality.
- Designed a user-friendly web interface where users could upload plant images and receive instant disease diagnosis and suggested remedies.
- Improved early disease detection for crops like tomato and potato, contributing to potential yield protection and farmer support.

Speech and music Classification using CNN | MATLAB, CNN, Audio Signal Processing

April 2024

- Designed and implemented a Convolutional Neural Network (CNN) model in MATLAB to accurately classify audio clips as speech or music.
- Preprocessed raw audio signals using techniques like MFCC (Mel Frequency Cepstral Coefficients) and spectrogram conversion to extract relevant features.
- Trained and validated the model on a custom-labeled dataset, achieving a classification accuracy of 88%.
- Visualized spectrograms and waveform patterns to understand key differences between speech and music segments.
- Automated audio classification workflow for real-time applications such as media tagging and voice/music separation.
- Enhanced model performance by tuning hyperparameters and implementing dropout layers to reduce overfitting.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL, AIML, Basic Java

Developer Tools: VS Code, Eclipse, MATLAB and Simulink, Android Studio, Cisco Packet Tracer, Jupyter

Technologies/Frameworks: JavaScript, React, GitHub, WordPress

Leadership / Extracurricular

National Service Scheme (NSS)

2022 – 2024

Coordinator

PCCOE

- Actively participate in national missions like **Swachh Bharat**, **Digital India**, or **Ek Bharat Shreshtha Bharat**.
- Skill development and training sessions for youth
- Road safety week, environment day, blood donation camp